Pseudocode and Logic for the DGSRHT Simulation Experiment

Explaining `run_experiment_DGSRHT` and the Four Plot Lines

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1 Overview

The simulation evaluates the **Distributed Sketching Efficiency** of two distributed OLS schemes—*DGSRHT* and *Uniform Sampling*—as a function of the number of working machines K'. For each scheme we plot two curves:

(i) An **empirical** curve: the Relative Efficiency (RE) estimated by averaging over R independent noise draws.

(ii) A **theoretical** curve: the RE predicted by the MSE formula $\text{MSE} = \sigma^2 \text{tr}\big((\mathbf{X}^\top \mathbf{X})^{-1}\big)$, applied to a single representative design matrix.

This gives the **four lines** in the plot.

2 Setup and Notation

Symbol	Meaning
n	Total number of observations across all k clients
k	Number of clients in the DGSRHT pipeline
p	Feature dimension
K'	Number of machines used in the *post-sketching* distributed OLS
s	Subsampling factor ($s = 1/\text{proportion}$)
R	Number of Monte Carlo repetitions (noise resamples)
$\boldsymbol{\beta}_{\text{true}}$	Ground-truth coefficient vector
Σ_1	Random SPD covariance for GMM component 1
$\Sigma_2 = c\,\Sigma_1$	Covariance for GMM component 2

Linear model. Each client i holds a fixed design matrix $\mathbf{X}_i \in \mathbb{R}^{m_i \times p}$ (drawn once from a two-component GMM). The response is:

$$\mathbf{Y}_i = \mathbf{X}_i \boldsymbol{\beta}_{\text{true}} + \boldsymbol{\varepsilon}_i, \qquad \boldsymbol{\varepsilon}_i \sim \mathcal{N}(\mathbf{0}, \mathbf{I}_{m_i}).$$

Crucially, $\mathbf{X}_i$ is **fixed** across all R repetitions; only$\boldsymbol{\varepsilon}_i$ is redrawn.

3 Full Pseudocode

Algorithm 1 RUN_EXPERIMENT_DGSRHT: Empirical + Theoretical RE

Input: $n, k, p, \text{weight}, \mu_1, \mu_2, c, s, K'_{\text{list}}, R, \text{base_seed}$

Phase A: Generate the fixed design (done once)

Draw $\boldsymbol{\beta}_{\text{true}} \sim \mathcal{N}(\mathbf{0}, \mathbf{I}_p)$
Draw random SPD covariance Σ_1; set $\Sigma_2 = c\,\Sigma_1$
Draw heterogeneous client sizes $(m_1, \ldots, m_k)$ via multinomial (with $\sum m_i = n$)
Compute $m_{\text{loc}} = \text{next_power_of_two}(\max_i m_i)$
for $i = 1, \ldots, k$ **do**
  Sample $\mathbf{X}_i^{\text{raw}} \in \mathbb{R}^{m_i \times p}$ from the two-component GMM
  Pad: $\mathbf{X}_i^{\text{pad}} = \begin{bmatrix} \mathbf{X}_i^{\text{raw}} \\ \mathbf{0}_{(m_{\text{loc}} - m_i) \times p} \end{bmatrix}$
end for

Phase B: Monte Carlo loop (only ε changes)

Initialize accumulators: $\text{MSE}_{\text{unif}}^{\text{OLS}}$, $\text{MSE}_{\text{DG}}^{\text{OLS}}$, $\text{MSE}_{\text{unif}}[K']$, $\text{MSE}_{\text{DG}}[K']$
for $r = 1, \ldots, R$ **do**
  % — Fresh noise —
  for $i = 1, \ldots, k$ **do**
    Draw $\boldsymbol{\varepsilon}_i^{(r)} \sim \mathcal{N}(\mathbf{0}, \mathbf{I}_{m_i})$
    $\mathbf{Y}_i^{\text{raw}} \leftarrow \mathbf{X}_i^{\text{raw}} \boldsymbol{\beta}_{\text{true}} + \boldsymbol{\varepsilon}_i^{(r)}$
    Pad $\mathbf{Y}_i^{\text{pad}}$ with zeros to length m_{loc}
  end for

  % — DGSRHT pipeline (produces sketched data) —
  $(\mathbf{X}^*, \mathbf{Y}^*) \leftarrow \text{DGSRHT}\big(\{\mathbf{X}_i^{\text{pad}}\}, \{\mathbf{Y}_i^{\text{pad}}\}, m_{\text{loc}}, s\big)$
    (local shuffle → local RHT → shared mask → server Hadamard mixing)

  % — Uniform baseline (same compression rate) —
  $(\mathbf{X}^{\text{unif}}, \mathbf{Y}^{\text{unif}}) \leftarrow \textsc{UniformSubsample}\big(\{\mathbf{X}_i^{\text{raw}}\}, \{\mathbf{Y}_i^{\text{raw}}\}, s\big)$

  % — Centralized OLS on each sketched dataset —
  $\hat{\boldsymbol{\beta}}_{\text{unif}}^{\text{OLS}} \leftarrow (\mathbf{X}^{\text{unif}\top}\mathbf{X}^{\text{unif}})^{-1}\mathbf{X}^{\text{unif}\top}\mathbf{Y}^{\text{unif}}$
  $\hat{\boldsymbol{\beta}}_{\text{DG}}^{\text{OLS}} \leftarrow (\mathbf{X}^{*\top}\mathbf{X}^*)^{-1}\mathbf{X}^{*\top}\mathbf{Y}^*$
  $\text{MSE}_{\text{unif}}^{\text{OLS}} \mathrel{+}= \|\hat{\boldsymbol{\beta}}_{\text{unif}}^{\text{OLS}} - \boldsymbol{\beta}_{\text{true}}\|^2$
  $\text{MSE}_{\text{DG}}^{\text{OLS}} \mathrel{+}= \|\hat{\boldsymbol{\beta}}_{\text{DG}}^{\text{OLS}} - \boldsymbol{\beta}_{\text{true}}\|^2$

  % — Distributed OLS for each K' —
  for each K' in K'_{list} **do**
    Randomly partition $(\mathbf{X}^{\text{unif}}, \mathbf{Y}^{\text{unif}})$ into K' blocks
    Compute local OLS on each block, average: $\bar{\boldsymbol{\beta}}_{K'}^{\text{unif}} = \frac{1}{K'}\sum_{j=1}^{K'} \hat{\boldsymbol{\beta}}_j$
    $\text{MSE}_{\text{unif}}[K'] \mathrel{+}= \|\bar{\boldsymbol{\beta}}_{K'}^{\text{unif}} - \boldsymbol{\beta}_{\text{true}}\|^2$

    Randomly partition $(\mathbf{X}^*, \mathbf{Y}^*)$ into K' blocks
    Compute local OLS on each block, average: $\bar{\boldsymbol{\beta}}_{K'}^{\text{DG}} = \frac{1}{K'}\sum_{j=1}^{K'} \hat{\boldsymbol{\beta}}_j$
    $\text{MSE}_{\text{DG}}[K'] \mathrel{+}= \|\bar{\boldsymbol{\beta}}_{K'}^{\text{DG}} - \boldsymbol{\beta}_{\text{true}}\|^2$
  end for
end for

4 What Each Line in the Plot Represents

Line	What it computes
1. Uniform empirical	$\mathrm{RE}^{\mathrm{emp}}_{\mathrm{unif}}(K') = \dfrac{\frac{1}{R}\sum_{r=1}^{R} \lVert \hat{\boldsymbol{\beta}}^{\mathrm{OLS}}_{\mathrm{unif},r} - \boldsymbol{\beta}_{\mathrm{true}} \rVert^2}{\frac{1}{R}\sum_{r=1}^{R} \lVert \bar{\boldsymbol{\beta}}^{\mathrm{unif}}_{K',r} - \boldsymbol{\beta}_{\mathrm{true}} \rVert^2}$ Numerator = average MSE of *centralized* OLS on the uniformly subsampled data. Denominator = average MSE of *distributed* OLS (averaged estimator across K' machines) on the same data.
2. Uniform theoretical	$\mathrm{RE}^{\mathrm{th}}_{\mathrm{unif}}(K') = \dfrac{\sigma^2 \operatorname{tr}\big((\mathbf{X}^{\mathrm{unif}\top}\mathbf{X}^{\mathrm{unif}})^{-1}\big)}{\sum_{j=1}^{K'} \frac{\sigma^2}{K'^2} \operatorname{tr}\big((\mathbf{X}_j^\top \mathbf{X}_j)^{-1}\big)}$ Uses the OLS variance formula applied to a *single* design matrix, partitioned once.
3. DGSRHT empirical	Same as Line 1, but the data is $(\mathbf{X}^*, \mathbf{Y}^*)$ produced by the DGSRHT pipeline (local shuffle + RHT + shared mask + server Hadamard mixing).
4. DGSRHT theoretical	Same as Line 2, but applied to the DGSRHT-transformed design $\mathbf{X}^*$.

Interpretation of RE.

$$\mathrm{RE}(K') = \frac{\mathrm{MSE}_{\mathrm{centralized}}}{\mathrm{MSE}_{\mathrm{distributed\ with}\ K'\ \mathrm{machines}}}$$

- RE = 1 means no efficiency loss from distributing.
- RE < 1 means the distributed estimator has *higher* MSE than centralized OLS (efficiency loss due to partitioning).
- Smaller RE = worse distributed performance.

As K' increases, more machines means smaller local datasets, which inflates the local OLS variance, so RE decreases.

5 Why the Theoretical Lines Are Not Perfectly Smooth

One may expect the theoretical curve to be a smooth, deterministic function of K'. In our implementation, **it is not**, for the following reason:

(a) The theoretical RE for each K' is computed by calling `Relative_Efficiency_theory`, which internally calls `partition_matrix_not_necessary_divisable`.

(b) That function **randomly shuffles** the row indices of the design matrix and splits them into K' blocks.

(c) Because this random shuffle is done **independently for each** K', each value $\mathrm{RE}^{\mathrm{th}}(K')$ is based on a *different* random partition.

(d) The MSE formula $\sum_{j=1}^{K'} \frac{1}{K'^2} \operatorname{tr}\big((\mathbf{X}_j^\top \mathbf{X}_j)^{-1}\big)$ is sensitive to the specific rows that land in each block. Different random partitions yield different block compositions, hence different MSE values.

In short: The theoretical line uses a **single random partition per** K' (not averaged over many partitions), and each K' draws its own independent partition. This introduces partition-to-partition variability, making the curve jagged rather than smooth.

How to make it smoother (optional). If a smooth theoretical curve is desired, one could average the theoretical RE over multiple random partitions for each K':

$$\overline{\mathrm{RE}}^{\mathrm{th}}(K') = \frac{1}{T}\sum_{t=1}^{T} \mathrm{RE}_t^{\mathrm{th}}(K'),$$

where each $\mathrm{RE}_t^{\mathrm{th}}$ uses a different random partition. With $T = 50$–100, the theoretical curve would become nearly smooth. This was not done in the current code to keep computation light, but it is a straightforward extension.

6 Summary of the Simulation Logic

1. **Fix X**: generate GMM design matrices for all k clients *once*.
2. **Resample** $\boldsymbol{\varepsilon}$: for $r = 1, \ldots, R$, draw fresh Gaussian noise and compute $\mathbf{Y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\varepsilon}$.
3. **Sketch**: apply the DGSRHT pipeline (or uniform subsampling) to get compressed data.
4. **Estimate**: run centralized OLS and distributed OLS (for each K') on the compressed data.
5. **Compute RE**: ratio of centralized MSE to distributed MSE, averaged over the R noise draws (empirical lines) or computed from the design-matrix trace formula (theoretical lines).

The key design choice is that **X is fixed and only** $\boldsymbol{\varepsilon}$ **varies**, which is the standard OLS simulation protocol: we estimate the *conditional* MSE $\mathbb{E}_{\varepsilon}[\|\hat{\boldsymbol{\beta}} - \boldsymbol{\beta}\|^2 \mid \mathbf{X}]$.